

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING

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Program	Cybersecurity Internship Program — NetworkWalks Academy
Date	18 September 2026
Modules Completed	W2-PM1 (Multiple Kali Linux Tools — Footprinting & Reconnaissance) W2-PM5 (Zenmap Scanning)
Client/Target	1. networkwalks.com (secured written permission via the internship program) 2. My own local LAN network
Permission secured from client?	Yes
Phases Covered	Phase 1: Reconnaissance & Footprinting — Completed Phase 2: Scanning & Network Discovery — Completed

1. Liability Disclaimer

I have performed these activities only on systems and domains for which I had secured permission through the NetworkWalks internship program, or on infrastructure I own myself. All material in this report is for education and research purposes only. Nothing here should be used to break the law. The instructor, the authors, and NetworkWalks are not responsible for how this knowledge is used. Every action taken is my own responsibility. Misuse can lead to criminal charges, heavy fines, loss of employment, and a permanent record. In most countries, unauthorized access is a crime even when nothing is damaged.

2. Introduction

This report covers footprinting the networkwalks.com domain using six Kali Linux tools (W2-PM1) and scanning my own local network with Zenmap (W2-PM5). One module covers the footprinting phase and the other covers the scanning phase, so together they show how an attacker moves from gathering public information to mapping live hosts on a network. All commands were run in Kali Linux (footprinting) and on a Windows PC with Zenmap installed (scanning). Every step below includes the exact command used, the result observed, and a short note on why the finding matters from an attacker's point of view.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux	Operating system used to run all reconnaissance tools
WHOIS	Find domain registration details (registrar, creation/expiry dates, name servers)
WhatWeb	Fingerprint web technologies (server, CMS, plugins, IP address)
Nslookup	Resolve the domain name to its IP address using DNS
curl -I	Read the HTTP response headers returned by the website
Wafw00f	Detect whether a Web Application Firewall protects the site
DNSRecon	Enumerate DNS records (SOA, NS, MX, TXT, SRV) for the domain
Windows CMD (ipconfig)	Identify local IP address, subnet mask, and default gateway
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts and MAC addresses

4. Activities Performed

4.1 Footprinting & Reconnaissance

I performed reconnaissance against the networkwalks.com domain using six Kali Linux tools: WHOIS, WhatWeb, Nslookup, curl, Wafw00f, and DNSRecon. Each tool was used to collect a different type of information about the target.

First, I ran WHOIS against the domain. The results showed the domain is registered through GoDaddy, was created on 6

November 2019, and is set to expire on 6 November 2027. The name servers returned were NS6135.HOSTGATOR.COM and NS6136.HOSTGATOR.COM, and the registrant's own identity is masked behind Domains By Proxy privacy protection.

I then ran WhatWeb against the site. The results identified WordPress 7.1 and WordPress Download Manager 3.3.58 as the CMS and plugin in use, alongside Bootstrap 7.1 and jQuery 3.7.1, the server IP 192.232.216.135, the Apache web server, and a publicly listed contact address (info@networkwalks.com).

Using Nslookup, I resolved the domain name to its IP address. The result confirmed 192.232.216.135, matching the IP surfaced independently by WhatWeb.

I used curl -I to inspect the HTTP response headers returned by the site. This returned a full set of headers, including a permissions-policy header, a link header exposing the WordPress REST API endpoints (/wp-json/ and /wp-json/wp/v2/pages/53), a referrer-policy header, and a session cookie named __wpdm_client.

Next, I used Wafw00f to check whether a Web Application Firewall was protecting the site. The result identified ModSecurity (SpiderLabs) as the WAF in front of the domain.

Finally, I used DNSRecon to enumerate the domain's DNS records. The results returned the SOA and NS records (both on Hostgator name servers), an MX record pointing to mail.networkwalks.com, an SPF TXT record authorizing Hostgator/WebsiteWelcome mail servers, a Google site-verification TXT record, and eight SRV records for autodiscover, all pointing to cPanel's email autodiscovery service.

4.2 Network Scanning with Zenmap

For the second activity, I used Zenmap to perform network discovery on my own local network. This required identifying my local IP address and subnet, discovering live hosts, identifying their IP and MAC addresses, and generating a network topology.

I first used the Windows ipconfig command to identify my local configuration. This returned an IPv4 address of 10.0.0.10, a subnet mask of 255.255.255.0, and a default gateway of 10.0.0.1, confirming my machine sits on the 10.0.0.0/24 network.

I then entered 10.0.0.0/24 into Zenmap's Target field and ran a Ping Scan (nmap -sn 10.0.0.0/24) to identify active hosts. The scan completed in 8.07 seconds and found 3 hosts up out of 256 addresses scanned:

- 10.0.0.1 — MAC 52:54:00:12:35:00
- 10.0.0.2 — MAC 08:00:27:8A:35:D2
- 10.0.0.10 — MAC 08-00-27-6B-59-2F

After completing the scan, I opened the Topology tab in Zenmap and enabled the legend. The topology showed all three hosts (10.0.0.1, 10.0.0.2, and localhost/10.0.0.10) as green nodes, meaning each has fewer than 3 open ports, connected to localhost with dashed lines indicating no traceroute hop information was available on this local segment.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network-scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Domain registration & hosting infrastructure exposed	WHOIS returned registrar (GoDaddy), creation date (2019), expiry date (2027), and name servers (NS6135/NS6136.HOSTGATOR.COM). Registrant identity itself is masked by Domains By Proxy.	Reveals hosting provider and DNS infrastructure, which can help an attacker plan further reconnaissance	● Low
2	Web technology & plugin versions exposed	WhatWeb identified WordPress 7.1 and WordPress Download Manager 3.3.58, plus Bootstrap 7.1 and jQuery 3.7.1	Attackers may cross-reference exact software versions against known CVEs to plan targeted exploitation	● Medium
3	Server IP address identifiable	Nslookup resolved networkwalks.com to 192.232.216.135	Confirms the hosting location of the web service and enables direct scanning of that host	● Low
4	HTTP headers & REST API endpoint exposed	curl -I returned full response headers, including a link header pointing to the WordPress REST API (/wp-json/ and /wp-json/wp/v2/pages/53) and a session cookie (__wpdm_client)	The exposed REST API endpoint could be enumerated for further information disclosure; header details assist fingerprinting	● Low
5	WAF technology identifiable	Wafw00f identified the site as protected by ModSecurity (SpiderLabs)	Knowing the specific WAF product lets an attacker research known bypass techniques for that WAF	● Low
6	DNS & mail infrastructure exposed	DNSRecon enumerated the SOA/NS records, an MX record (mail.networkwalks.com), an SPF TXT record, a Google site-verification TXT record, and SRV records for autodiscover pointing to cPanel email discovery servers	Provides a fuller map of the mail and hosting infrastructure, useful to an attacker building an infrastructure profile or planning phishing/spoofing attempts	● Medium
7	Live hosts identifiable on local	Zenmap's ping scan identified 3 live hosts with their MAC addresses on the subnet in 8	An attacker on the same network segment could use this to quickly map active	● Medium

network	seconds	devices and target the gateway or other hosts directly
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Risk level key: ● Critical ● Medium ● Low

These risks are observations from the footprinting and scanning exercises, not confirmed vulnerabilities. Only publicly available information was gathered and only a ping scan was run on the local network; no exploitation or vulnerability validation was performed as part of either module. The presence of a software version, IP address, DNS record, or live host does not by itself mean the system is vulnerable, further authorized testing would be required to confirm any actual weakness.

6. Recommendations

Review publicly exposed technology information The organization should regularly review what its WhatWeb/HTTP-header footprint reveals about its CMS, plugins, and server stack, and remove anything not needed publicly.

Keep CMS and plugins updated WordPress and WordPress Download Manager should be kept current and checked against published security advisories, since exact version numbers are already visible to anyone running WhatWeb.

Review HTTP response headers Headers exposing internal REST API paths (e.g. /wp-json/wp/v2/pages/53) and cookie names should be reviewed to ensure only necessary information is disclosed.

Review DNS records periodically SOA, MX, TXT, and SRV records should be audited regularly so that only records genuinely required for mail and hosting remain publicly resolvable.

Keep the WAF properly configured and monitored ModSecurity is already in place and should continue to be tuned and monitored, since it currently blocks naive automated attacks.

Maintain registrar privacy protection Domain privacy protection (Domains By Proxy) is already enabled and should be kept in place to avoid exposing registrant contact details.

Perform regular internal network discovery Since a simple ping scan located every device on my /24 network in under 10 seconds, the same network should be periodically scanned by its owner to confirm only expected devices (gateway, VMs) are present.

Perform security testing only with authorization Any further reconnaissance, scanning, or testing against this domain or network should continue to be carried out strictly within the permission granted by the internship program or my own ownership of the network.

7. Conclusion

During Week 2 of my Cybersecurity internship with NetworkWalks Academy, I completed practical activities covering footprinting, reconnaissance, and network scanning.

In the footprinting activity, I used six Kali Linux tools to collect information about the target domain. I learned how WHOIS reveals domain registration and hosting details, WhatWeb fingerprints web technologies and plugin versions, Nslookup resolves a domain to its IP address, curl exposes HTTP response headers and hidden API endpoints, Wafw00f identifies the WAF protecting a site, and DNSRecon enumerates a domain's DNS, mail, and service records.

In the network-scanning activity, I used Zenmap to identify my local network configuration and discover active hosts. My /24 network had only 3 live hosts, my default gateway, an Oracle VirtualBox VM, and my own machine — and I generated a topology diagram confirming this layout.

These exercises showed me how much information can be gathered both about a public target and a private network before any exploitation begins, simply by querying public records, the server's own responses, and running a basic ping sweep. A good cybersecurity report should explain what was performed, what was discovered, what the finding means, what risk it may create, and what can be done to reduce that risk.

Finally, reconnaissance and scanning must always be performed within an authorized scope, these activities were completed as part of the assigned educational cybersecurity lab.

8. Evidences Collected

Terminal output for each command (whois, whatweb, nslookup, curl -I, wafw00f, dnsrecon) was captured during the footprinting exercise. For the scanning exercise, screenshots of the Zenmap Nmap Output (ping scan results), the ipconfig command output, and the Zenmap Topology view with legend enabled were captured and are retained alongside this report as supporting evidence.

